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Protocol for updated systematic review and meta-analysis on the burden of non-communicable diseases among people living with HIV in Sub-Saharan Africa

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Manuscripts

Title: Protocol for updated systematic review and meta-analysis on the burden of non-communicable diseases among people living with HIV in Sub-Saharan Africa

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Abstract (299 words)

Introduction: Sub Saharan Africa (SSA) is faced with the dual epidemics of HIV/AIDS and Non-Communicable Diseases (NCDs). Cardiovascular diseases, cancers, chronic respiratory diseases, diabetes, and mental illnesses are the five major NCDs, causing death globally with low-and medium-income countries, contributing 78% of all NCD deaths and 85% of premature deaths. There has been increased interest in the integration of HIV and NCDs care, especially in SSA that accounts for 55% of people living with HIV (PLHIV) globally. This systematic review and meta-analysis will determine the overall prevalence or incidence of NCDs (or its risk factors) among adults living with in HIV in SSA.

Methods and Analysis: The Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines will be utilised. The title and abstracts of the identified observational studies published since 2010 will be screened for mention of HIV and any NCD/NCD risk factor. Study participants will be any adult (≥ 18 years old) living with HIV in SSA. Exposures of interest will include any NCD/NCD risk factors and ART. Outcomes of interest are prevalence or incidence of any NCD/NCD risk factors. A random-effects meta-analysis will be used to estimate pooled prevalence or incidence of the five major NCDs among PLHIV, using Stata software. Chi-square test and I^2 statistic will be used to measure statistical heterogeneity between studies. If there is significant heterogeneity, subgroup analysis will be used to investigate potential sources. Publication bias will be assessed using funnel plots and the Stata "**metabias"** command.

Ethics and Dissemination: Ethical review will not be required because it is a systematic review. Data will be kept in the institutional data repository. Study findings will be disseminated through peer reviewed publications and conference presentations.

The study protocol is registered in the international prospective register of systematic reviews (https://www.crd.york.ac.uk/PROSPERO/), registration number is CRD42021258769.

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Keywords: Cancer, Cardiovascular Diseases, Chronic Respiratory Diseases, Diabetes, HIV/AIDS, Non-Communicable Diseases, Hypertension, Mental Illnesses, Sub-Saharan Africa

Strengths and Limitations of this study:

- The study aims to report the current burden of the five major NCDs and their respective risk factors among PLHIV in SSA.
- Peer reviewed studies published since 2010 will be reviewed.
- Pubmed/Medline, EMBASE, Google scholar databases will be used to search for indexed publications.
- Increased heterogeneity for some of the NCDs/risk factors may not allow for meta-analyses.

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Introduction (2422 words)

Sub Saharan Africa (SSA) is faced with the syndemic of HIV and non-communicable diseases (NCDs) that have been widely reported to be on the increase globally(1-7). In 2016, NCDs accounted for 41 million deaths annually (71% of all deaths globally),(8) with over 85% of premature deaths (between the ages of 30 and 69 years) from NCDs, occurring in low-and middle-income countries (LMICs)(2, 4).

Cardiovascular diseases (CVDs), cancers, chronic respiratory diseases (CRDs) and diabetes are historically the four main groups of diseases that account for over 80% of all NCD related deaths(3, 4, 9). However, the World Health Organization (WHO) has recently classified mental illnesses as one of the main NCDs(7). Moreover, in 2016, suicide was responsible for 800,000 deaths globally(7). NCDs are on the increase in LMICs that are also burdened with multiple infectious diseases (6, 10, 11), including COVID-19 pandemic which is a current challenge. This increase in NCDs has been attributed to the implementation of universal testing and treatment that has led to increased coverage of antiretroviral treatment (ART) which is enabling people living with HIV (PLHIV) to live a relatively normal lifespan, resulting in increased probability of NCD related mortality being more than that of HIV/AIDS-related deaths.

Risk factors for NCDs have been previously classified into “Metabolic” and “Modifiable behavioural ” risk factors(3). Metabolic risk factors include raised blood pressure, overweight/obesity, hyperglycemia and hyperlipidaemia(3) while the modifiable behavioural risk factors include tobacco use (including second-hand smoke), physical inactivity, unhealthy diet, and the harmful use of alcohol. Environmental air pollution has also been identified as a key risk factor for NCDs in general (7).

Daily tobacco smoking has been associated with several morbidities, including CVDs and lung cancer(12). Air pollution is also a major risk factor for NCD comparable to current

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tobacco use and is responsible for millions of deaths from ischemic heart disease, chronic lung diseases and cancers(7). Physical inactivity is also associated with CVD, type 2 diabetes, hypertension, obesity and high low density lipoprotein (LDL) cholesterol and certain cancers such as colon and breast cancer (13). Unhealthy diet is associated with obesity, increased risk of hypertension, diabetes, CVDs, cancers and CRDs. The harmful use of alcohol is also associated with CVDs, cancers, and liver diseases(3, 5).

NCDs have been reported to be a challenge to economic development (3, 7, 14). Globally, 37 % of the deaths occur in people aged 30-69 years old, which is the working age group. Employers face high staff turnover due to prolonged absenteeism and eventually deaths. Household costs associated with health care, increase in families that are already faced with financial difficulties in LMICs due to the chronic nature of the disease(3, 14). NCDs tend to foster poverty as usually it is breadwinners whose lives are lost, thus leaving families in financial difficulties, especially in LMICs. According to WHO, NCDs threaten progress towards the 2030 Agenda for Sustainable Development, that includes a target of reducing premature deaths from NCDs by one-third by 2030 (3, 9). For example, in South Africa, NCDs remain the major cause of death in the young working population and was previously reported to have a loss of US$1.88 billion from its gross domestic product due to management of NCDs between 2006 and 2015(14).

Among PLHIV, NCDs have equally been documented to be on the increase (2, 5, 15, 16). It has previously been reported that PLHIV are at risk of contracting NCDs from HIV infection itself, from ART and from the risk associated with increasing age. Individuals on ART are now able to live longer and therefore have an increased risk of chronic co-morbidities, including CVD, diabetes, cancers, lipodystrophy, and metabolic abnormalities(5, 15, 17). SSA accounts for 55% of the 38 million PLHIV (18) globally. Therefore, it is imperative that NCDs among PLHIV are investigated and that public health systems in SSA implement

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integrated NCD/HIV care for PLHIV, allowing for comprehensive health care provision. Patel et al (2018) is a published systematic review on the burden of NCDs among PLHIV, however it has not been updated and this study will seek to update the evidence from 2010 to the present date of search and this will also include chronic respiratory diseases that were not included in this publication(5). A systematic review and meta-analysis will be conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) of 2020(19).

AIM

The aim of the systematic review and meta-analysis is to determine the burden of NCDs and NCD risk factors among PLHIV in SSA.

Objectives

1. To determine the prevalence or incidence of any NCD (5 main NCDs) among PLHIV in SSA
2. To determine the prevalence or incidence of risk factors of any NCD among PLHIV in SSA
3. To determine if there is an association between being on ART and NCDs and/or NCD risk factors among PLHIV in SSA.
4. To determine if duration on ART is associated with NCDs and/or NCD risk factors among PLHIV in SSA.

METHODS

Eligibility Criteria

Any study that focuses on the burden of any of the five major NCDs and their respective risk factors among PLHIV in SSA will be eligible. The title and abstracts of the identified articles will be screened for mention of HIV and any NCD; CVD risk factors (e.g., hypertension,

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dyslipidaemia, hyperlipidaemia, and obesity), cancer, diabetes, mental illnesses and chronic respiratory disease.

Study participants will be any adult (≥ 18 years old) living with HIV in SSA. Exposures of interest will include any NCD, NCD risk factors and ART. Outcomes of interest are prevalence or incidence of any NCD or NCD risk factors. Study designs that will be reviewed are observational studies (cross sectional and cohort), HIV and NCD reports, systematic reviews, Demographic Health Surveys (DHS) and other similar studies.

Information Sources

The electronic databases that will be searched for eligible peer reviewed articles are PubMed/MEDLINE, EMBASE and Google scholar online databases. A PubMed/MEDLINE search strategy will be developed and adapted for all other databases. Boolean operators, Medical Subject Heading (MeSH) terms and key words will be used as part of the search strategy. Where necessary, study authors will be contacted for verification of published data.

Grey literature will also be searched for relevant studies. A hand search for key HIV/AIDS and NCD journals will be conducted.

Bibliographies of the included studies will be checked as a measure to identify further eligible studies. Included studies will be corroborated with those included in the previously published systematic reviews.

All studies investigating NCDs and their associated risk factors among PLHIV since 2010 will be included. HIV studies that do not include any of the five main NCDs will be excluded. Clinical trials and systematic reviews will be included for bibliographic searches but will be excluded for the meta-analysis. Studies within the scope of this study that were published before 1st January 2010 will be excluded. The date of the last search will be documented.

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Outcomes

Primary outcomes

- The prevalence or incidence of any NCD (five main NCDs) among PLHIV in SSA
- The prevalence or incidence of NCD risk factors for any NCD among PLHIV in SSA

Secondary Outcomes

- Odds ratios or Risk ratios for association of exposure to ART and NCDs and NCDs risk factors among PLHIV in SSA.

The keywords to be used in the search are HIV/AIDS, Non-Communicable Diseases, Cardiovascular Disease, Hypertension, Hyperlipidaemia, Dyslipidaemia, Diabetes, Cancer, Cervical cancer, Mental Illnesses, Depression, Chronic Respiratory Diseases and Asthma

Search Strategy

A modified search strategy of Patel et al (5) will be utilised. Below is Pubmed/Medline search strategy for "cardiovascular disease" as indicated by Patel (2018). A similar strategy will be employed for the other 4 NCDs and risk factors of interest.

Cardiovascular (PLWH* OR PLWHA* OR "people living with HIV" OR "people living with HIV/AIDS" OR "people living with" OR "people living with aids" OR "people living with hiv" OR "people living with hiv/aids" OR "people living with hiv/aids plwha" OR "people living with hiv/aids plwhas" OR "people living with hiv aids" OR "people living with hiv aids plwha" OR "people living with hiv aids plwhas" OR "people living with hiv and aids" OR "people living with hiv infection" OR "people living with hivs" OR "people living with human immunodeficiency virus" OR "people living with human immunodeficiency virus/acquired immunodeficiency syndrome" OR "people living with human immunodeficiency virus acquired immunodeficiency

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syndrome" OR "people living with hiv" OR “HIV Infections”[mesh] OR “HIV infection” OR “HIV infections” OR PLWHIV OR “HIV positive” OR “HIV-positive” OR “HIV+” OR “HIV infected” OR “HIV-infected” OR “HIV seropositivity”[mesh] OR “HIV seropositivity”) AND (LMIC OR LMICs OR "low- and middle- income countries" OR "low- and middle- income country" OR "low and middle income" OR "low- and middle- income" OR "lower middle income" OR "lmics" OR "lower middle income" OR "lower middle income countries" OR "lower middle income country" OR "lower middle income economies" OR "lower middle income setting" OR "lower middle income settings" OR Cote d’Ivoire[mesh] OR Ethiopia[mesh] OR Kenya[mesh] OR Malawi[mesh] OR South Africa[mesh] OR Uganda[mesh] OR Zambia[mesh] OR Ivory Coast[tiab] OR Cote d’Ivoire[tiab] OR Ethiopia[tiab] OR Kenya[tiab] OR Malawi[tiab] OR South Africa[tiab] OR Uganda[tiab] OR Zambia[tiab] OR "sub Saharan Africa" OR "sub-saharan Africa" OR Africa[tiab] OR Africa[mesh] OR “sub saharan Africa” OR “developing country” OR “developing countries” OR Cameroon[tiab] OR Central African Republic[tiab] OR Chad[tiab] OR Congo[tiab] OR Equatorial Guinea[tiab] OR Gabon[tiab] OR Democratic Republic of the Congo[tiab] OR Burundi[tiab] OR Djibouti[tiab] OR Ethiopia[tiab] OR Kenya[tiab] OR Rwanda[tiab] OR Somalia[tiab] OR Sudan[tiab] OR Tanzania[tiab] OR Uganda[tiab] OR Angola[tiab] OR Botswana[tiab] OR Lesotho[tiab] OR Malawi[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR South Africa[tiab] OR Swaziland[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Benin[tiab] OR Burkina Faso[tiab] OR Cote D'ivoire OR Gambia OR Ghana OR Guinea OR Guinea-Bissau OR Liberia OR Mali OR Mauritania OR Niger[tiab] OR Nigeria[tiab] OR Senegal[tiab] OR Sierra Leone[tiab] OR Togo[tiab] OR Cameroon[mesh] OR Central African Republic[mesh] OR Chad[mesh] OR Congo[mesh] OR Equatorial

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Guinea[mesh] OR Gabon[mesh] OR Democratic Republic of the Congo[mesh] OR Burundi[mesh] OR Djibouti[mesh] OR Ethiopia[mesh] OR Kenya[mesh] OR Rwanda[mesh] OR Somalia[mesh] OR Sudan[mesh] OR Tanzania[mesh] OR Uganda[mesh] OR Angola[mesh] OR Botswana[mesh] OR Lesotho[mesh] OR Malawi[mesh] OR Mozambique[mesh] OR Namibia[mesh] OR South Africa[mesh] OR Swaziland[mesh] OR Zambia[mesh] OR Zimbabwe[mesh] OR Benin[mesh] OR Burkina Faso[mesh] OR Cote D'ivoire[mesh] OR Gambia[mesh] OR Ghana[mesh] OR Guinea[mesh] OR Guinea-Bissau[mesh] OR Liberia[mesh] OR Mali[mesh] OR Mauritania[mesh] OR Niger[mesh] OR Nigeria[mesh] OR Senegal[mesh] OR Sierra Leone[mesh] OR Togo[mesh] OR "Africa, Central"[Mesh] OR "Africa, Eastern"[Mesh] OR "Africa, Southern"[Mesh] OR "Africa, Western"[Mesh]) AND (Cardiovascular Diseases[mesh] OR Heart Diseases[mesh] OR hypertension[mesh] OR stroke[mesh] OR cardiovascular OR "heart disease" OR hypertension OR "high blood pressure" OR stroke OR "heart attack" OR "Coronary Disease"[Mesh] OR "Cerebrovascular Disorders"[Mesh] OR "Pulmonary Embolism"[Mesh] OR "Peripheral Arterial Disease"[Mesh] OR "Peripheral Vascular Diseases"[Mesh] OR "Rheumatic Heart Disease"[Mesh] OR "Venous Thrombosis"[Mesh] OR "coronary disease" OR "pulmonary embolism" OR "cerebrovascular disorder" OR "cerebrovascular disease" OR "peripheral arterial disease" OR "rheumatic heart disease" OR "deep vein thrombosis" OR “ischemic heart disease” OR “heart failure” OR “coronary heart disease” OR “cardiovascular disease” OR “Inflammation"[Mesh] OR inflammation OR inflame* OR "Atherosclerosis"[Mesh] OR atherosclerosis OR "Metabolic Syndrome X"[Mesh] OR “metabolic cardiovascular syndrome” OR “metabolic syndrome” OR “metabolic syndrome X” OR “insulin resistance syndrome X” OR “metabolic X syndrome” OR “cardiovascular biomarker” OR “inflammation

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biomarker" OR "endothelial function" OR "Interleukin-6"[Mesh] OR IL-6 OR "C-Reactive Protein"[Mesh] OR "C Reactive Protein" OR "C-Reactive Protein" OR "Carotid Intima-Media Thickness"[Mesh] OR "carotid intima media thickness" OR "carotid intima-media thickness" OR "Cholesterol"[Mesh] OR cholesterol OR "Angiography"[Mesh] OR angiography OR "HydroxymethylglutarylCoA Reductase Inhibitors"[Mesh] OR "Hydroxymethylglutaryl CoA Reductase Inhibitors" OR "HMG-CoA Statins" OR "HMG-CoA Reductase Inhibitors" OR "Fluorodeoxyglucose F18"[Mesh] OR 18F-FDG OR "18F FDG" OR "Fluorodeoxyglucose F 18" OR "2-Fluoro-2-deoxy-D-glucose" OR "Venous Thromboembolism"[Mesh] OR "venous thromboembolism" OR cardiometabolic OR "Myocardial Infarction"[Mesh] OR "myocardial infarction" OR "myocardial infarct" OR "Vascular Diseases"[Mesh] OR "vascular disease" OR "Coronary Artery Disease"[Mesh] OR "coronary artery disease" OR "Myocarditis"[Mesh] OR myocarditis OR "Cardiomyopathies"[Mesh] OR cardiomyopathy OR cardiomyopathies OR "cardiac disease" OR "cardiac arrhythmias" OR "Arrhythmias, Cardiac"[Mesh] OR arrhythmia* OR "myocardial disease" OR "myocardial diseases" OR myocardiopathy OR myocardiopathies OR carditis OR "dyslipidemias"[MeSH] OR dyslipidemia OR hyperlipidemia OR hypercholesterolemia OR hypertriglyceridemia OR triglyceride OR triglycerides OR HDL OR LDL OR VLDL OR "Lipoproteins, HDL"[Mesh] OR "Lipoproteins, LDL"[Mesh] OR "Lipoproteins, VLDL"[Mesh] OR hyperlipoproteinemia OR lipoprotein(a) OR hyperlipidaemia OR hypercholesterolaemia OR hypertriglyceridaemia OR "Blood Pressure"[MeSH] OR "blood pressure" OR "systolic blood pressure" OR "diastolic blood pressure" OR SBP[tiab] OR DBP[tiab])

Study Selection

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Two independent review authors will screen the potentially relevant titles and abstracts according to pre specified eligibility criteria. Selected articles from the respective databases will be transferred to EndNoteX9 (https://endnote.com/). These articles will then be transferred to Rayyan software (https://www.rayyan.ai/) for screening of titles and abstracts. Full text articles will be retrieved after screening the titles and abstracts. Any disagreements arising from record screening will be resolved through discussion.

Data Extraction

A pre-designed data extraction form will be used by two review authors who will independently extract data on prevalence or incidence of any NCD (cardiovascular disease, cancer, mental illness, chronic respiratory disease and diabetes) among PLHIV, prevalence or incidence of NCD risk factors (hypertension, hyperlipidaemia, dyslipidaemia, physical inactivity, obesity, smoking and pollution) in PLHIV, study design, sample size, participants' age, recruitment methods, study country and date of study publication. The same review authors will independently assess each included study for risk of bias with respect to sequence generation, incomplete outcome data, selective reporting, and other potential sources of bias.

Discrepancies will be resolved by discussion or by consulting with a third review author. Data will be transferred to Stata version 16 (Stata IC/16.0, StataCorp LLC).

Missing Data

Study authors will be contacted concerning missing data on either outcomes or risk of bias.

Data Synthesis

Meta-analysis will be performed using the **"metan"** and **"metaprop"** commands in Stata version 16. A random-effects meta-analysis will be used to estimate pooled prevalence or incidence of the 5 major NCDs among PLHIV. The pooled prevalence or incidence of

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hypertension, hyperlipidaemia, obesity, overweight and waist circumference in addition to the prevalence of modifiable risk factors (tobacco use, alcohol intake, diet, and physical activity) will be determined. The main outcome of interest will be the prevalence of NCD and NCD risk factors amongst PLHIV with respective 95% confidence intervals. Forest plots will be constructed to display meta-analysis results. Chi-square test and I^2 statistic will be used to measure statistical heterogeneity between studies. If there is significant heterogeneity, subgroup analysis will be used to investigate potential sources; this may include the type of NCD, country, or gender (Male/Female). Publication bias will be assessed using funnel plots and the Stata **"metabias"** command.

Patient and public Involvement

No patient involved.

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Author Contribution: MM -Conception of the study, Writing - original draft and editing; AM - Conception and design of the study, Writing- reviewing and editing, Supervision. All authors approved the final version to be submitted.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review
Update	1b	If the protocol is for an update of a previous systematic review, identify as such
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments
Support:		
Sources	5a	Indicate sources of financial or other support for the review
Sponsor	5b	Provide name for the review funder and/or sponsor
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated
Study records:		
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review

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Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Protocol for updated systematic review and meta-analysis on the burden of non-communicable diseases among people living with HIV in Sub-Saharan Africa

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Title: Protocol for updated systematic review and meta-analysis on the burden of non-communicable diseases among people living with HIV in Sub-Saharan Africa

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Abstract (292words)

Introduction: Sub Saharan Africa (SSA) is faced with the dual epidemics of HIV/AIDS and Non-Communicable Diseases (NCDs). Cardiovascular diseases, cancers, chronic respiratory diseases, diabetes, and mental illnesses are the five major NCDs, causing death globally with low-and middle-income countries, contributing 78% of all NCD deaths and 85% of premature deaths. There has been increased interest in the integration of HIV and NCDs care, especially in SSA that accounts for 55% of people living with HIV (PLHIV) globally. This systematic review and meta-analysis will estimate the overall prevalence or incidence of NCDs (or its risk factors) among adults living with HIV in SSA.

Methods and Analysis: The Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines will be utilised. Two authors will independently screen the title and abstracts of the articles identified from the search. Study participants will be any adult (≥ 18 years old) living with HIV in SSA. Exposure of interest will be HIV (with or without ART). Outcomes of interest are prevalence or incidence of any NCD/NCD risk factors. A random-effects meta-analysis will be used to estimate pooled prevalence or incidence of the five major NCDs among PLHIV, using Stata software. Chi-square test and I^2 statistic will be used to measure statistical heterogeneity between studies. If there is significant heterogeneity, subgroup analysis will be used to investigate potential sources. Publication bias will be assessed using funnel plots and the Stata "**metabias"** command.

Ethics and Dissemination: Ethical review will not be required because it is a systematic review. Data will be kept in the institutional data repository. Study findings will be disseminated through peer reviewed publications and conference presentations.

The study protocol is registered in the international prospective register of systematic reviews (https://www.crd.york.ac.uk/PROSPERO/), registration number is CRD42021258769.

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Keywords: Cancer, Cardiovascular Diseases, Chronic Respiratory Diseases, Diabetes, HIV/AIDS, Non-Communicable Diseases, Hypertension, Mental Illnesses, Sub-Saharan Africa

Strengths and Limitations of this study:

- The study aims to report the current burden of the five major NCDs and their respective risk factors among PLHIV in SSA.
- Peer reviewed studies published since 2010 will be reviewed.
- Pubmed/Medline, EBSCOhost and Scopus databases will be used to search for indexed publications.
- Increased heterogeneity for some of the NCDs/risk factors may not allow for meta-analyses.

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Introduction (2662 words)

Sub Saharan Africa (SSA) is faced with the syndemic of HIV and non-communicable diseases (NCDs) that have been widely reported to be on the increase globally[1-4]. In 2016, NCDs accounted for 41 million deaths annually (71% of all deaths globally),[3] with over 85% of premature deaths (between the ages of 30 and 69 years) from NCDs, occurring in low-and middle-income countries (LMICs)[3 5].

Cardiovascular diseases (CVDs), cancers, chronic respiratory diseases (CRDs) and diabetes are historically the four main groups of diseases that account for over 80% of all NCD related deaths[3 6 7]. However, the World Health Organization (WHO) has recently classified mental illnesses as one of the main NCDs[8]. Moreover, in 2016, suicide was responsible for 800,000 deaths globally[8]. The NCDs are on the increase in LMICs that are also burdened with infectious diseases [9-11], including COVID-19 pandemic, which is a current challenge. Furthermore, WHO has reported disruptions in NCD services due to the COVID-19 pandemic globally, including in 41 African countries. The main reasons for the disruption to the NCD services are the decrease in inpatient volume due to cancellation of elective services, closure of population level screening, Government or public transport lock-downs hindering access to the health facilities for patients, NCD related clinical staff deployed to provide COVID-19 relief, closure of outpatient disease specific consultation clinics, insufficient personal protective equipment available for health care providers to provide services, insufficient staff to provide services, closure of outpatient NCD services as per government directive, decrease in outpatient volume due to patients not presenting, and inpatient services/hospital beds not available and stock out of essential medicines, medical diagnostics or other health products at health facilities[12]. This disruption in NCD services has highlighted the need for poor resourced settings such as SSA to strengthen their health systems, also as a means of improving preparedness for future epidemics. Moreover, patients

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with pre-existing co-morbidities were at a higher risk of severe disease due to the virus, and hence a need to ensure that NCD patients are well managed also as part of strengthening the health systems.

The increase in NCDs has been attributed to physical inactivity, unhealthy diets, harmful use of alcohol and tobacco in LMICs, just as in the developed countries [3]. Lifestyle changes in SSA due to the growth of urbanisation, that entails more sedentary life style as urban work is often less physical, exposure to unhealthy diets that are high in salt, fat and sugar and pollution, among other factors have contributed to the NCD burden[13]. Among PLHIV, there have been reports that HIV itself or ART side effects predispose individuals further to NCDs[14]. In addition, the general increase in lifespan implies increased risk to age-related NCDs.

Modifiable behavioural risk factors for NCDs include, tobacco use (including second-hand smoke), physical inactivity, unhealthy diet, and the harmful use of alcohol which consequently lead to raised blood pressure, overweight/obesity, hyperglycemia and hyperlipidaemia[7]. Environmental air pollution has also been identified as a key risk factor for NCDs in general [8].

Daily tobacco smoking has been associated with several morbidities, including CVDs and lung cancer[15]. Air pollution is also a major risk factor for NCD comparable to current tobacco use and is responsible for millions of deaths from ischemic heart disease, chronic lung diseases and cancers[8]. Physical inactivity is also associated with CVD, type 2 diabetes, hypertension, obesity and high low density lipoprotein (LDL) cholesterol and certain cancers such as colon and breast cancer [16]. Unhealthy diet is associated with obesity, increased risk of hypertension, diabetes, CVDs, cancers and CRDs. The harmful use of alcohol is also associated with CVDs, cancers, and liver diseases[2 7].

The NCDs are a challenge to economic development [7 8 17]. Globally, 37 % of the deaths occur

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in people aged 30-69 years old, which is the working age group. As a result, employers face high staff turnover due to prolonged absenteeism and eventually deaths. Additionally, household costs associated with health care, increase in families that are already faced with financial difficulties in LMICs due to the chronic nature of the disease[7 17]. Indeed, NCDs tend to foster poverty as usually it is breadwinners whose lives are lost, thus leaving families in financial difficulties, especially in LMICs. According to WHO, NCDs threaten progress towards the 2030 Agenda for Sustainable Development, that includes a target of reducing premature deaths from NCDs by one-third by 2030 [6 7]. For example, in South Africa, NCDs remain the major cause of death in the young working population and was previously reported to have a loss of US$1.88 billion from its gross domestic product due to management of NCDs between 2006 and 2015[17].

Among PLHIV, NCDs have equally been documented to be on the increase [1 2 18 19]. It has previously been reported that PLHIV are at risk of having NCDs from HIV infection itself, from ART and from the risk associated with increasing age. Individuals on ART are now able to live longer and therefore have an increased risk of chronic co-morbidities, including CVD, diabetes, cancers, lipodystrophy, and metabolic abnormalities[2 18 20]. SSA accounts for 55% of the 38 million PLHIV [21] globally. Therefore, it is imperative that NCDs among PLHIV are investigated and that public health systems in SSA implement integrated NCD/HIV care for PLHIV, allowing for comprehensive health care provision. Patel et al (2018)[2], a systematic review, included articles published between 2010 and 2016 on the burden of NCDs among PLHIV in LMICs. However, it has not been updated and there have been several publications between 2017 and 2021. This systematic review will update the evidence from 2010 to the present date and report on changes in trends of NCD burden in the HIV population. The updated systematic review will include chronic respiratory diseases that were exclude in the previous systematic review[2]. The systematic review and meta-analysis will be conducted

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using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) of 2020[22].

AIM

The aim of the systematic review and meta-analysis is to determine the burden of NCDs and NCD risk factors among PLHIV in SSA.

Objectives

1. To determine the prevalence or incidence of any NCD (5 main NCDs) among PLHIV in SSA
2. To determine the prevalence or incidence of risk factors of any NCD among PLHIV in SSA
3. To determine if there is an association between being on ART and NCDs and/or NCD risk factors among PLHIV in SSA.

METHODS

Eligibility Criteria

Any study published between 2010 to the current date, that focuses on the burden of any of the five major NCDs and their respective risk factors among PLHIV in SSA will be eligible. Two authors will independently screen the title and abstracts of the articles identified from the search .Study participants will be any adult (≥ 18 years old) living with HIV in SSA. The main exposure of interest will be HIV. Outcomes of interest are prevalence or incidence of any NCD or NCD risk factors in HIV populations. Study designs that will be reviewed are observational studies (cross sectional and cohort), HIV and NCD reports, systematic reviews, Demographic Health Surveys (DHS) and other similar studies.

Information Sources

The electronic databases that will be searched for eligible peer reviewed articles are PubMed/MEDLINE, Scopus and EBSCOhost online databases. A PubMed/MEDLINE

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search strategy will be developed and adapted for all other databases. Boolean operators, Medical Subject Heading (MeSH) terms and key words will be used as part of the search strategy. Where necessary, study authors will be contacted for verification of published data.

Grey literature will also be searched for relevant studies. A hand search for key HIV/AIDS and NCD journals will be conducted.

Bibliographies of the included studies will be checked as a measure to identify further eligible studies. Included studies will be corroborated with those included in the previously published systematic reviews.

All studies investigating NCDs and their associated risk factors among PLHIV since 2010 will be included. HIV studies that do not include any of the five main NCDs will be excluded. Clinical trials and systematic reviews will be included for bibliographic searches but will be excluded for the meta-analysis. Studies within the scope of this study that were published before 1st January 2010 will be excluded. The date of the last search will be documented.

Outcomes

Primary outcomes

- The prevalence or incidence of any NCD (five main NCDs) among PLHIV in SSA
- The prevalence or incidence of NCD risk factors for any NCD among PLHIV in SSA

Secondary Outcomes

- Odds ratios or Risk ratios for association of exposure to ART and NCDs and NCDs risk factors among PLHIV in SSA.

The keywords to be used in the search are HIV/AIDS, Non-Communicable Diseases, Cardiovascular Disease, Hypertension, Hyperlipidaemia, Dyslipidaemia, Diabetes,

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Cancer, Cervical cancer, Mental Illnesses, Depression, Chronic Respiratory Diseases and Asthma

Search Strategy

A modified search strategy of Patel et al [2] will be utilised. Below is Pubmed/Medline search strategy for "cardiovascular disease" as indicated by Patel (2018). A similar strategy will be employed for the other four NCDs and risk factors of interest.

Cardiovascular (PLWH* OR PLWHA* OR "people living with HIV" OR "people living with HIV/AIDS" OR "people living with" OR "people living with aids" OR "people living with hiv" OR "people living with hiv/aids" OR "people living with hiv/aids plwha" OR "people living with hiv/aids plwhas" OR "people living with hiv aids" OR "people living with hiv aids plwha" OR "people living with hiv aids plwhas" OR "people living with hiv and aids" OR "people living with hiv infection" OR "people living with hivs" OR "people living with human immunodeficiency virus" OR "people living with human immunodeficiency virus/acquired immunodeficiency syndrome" OR "people living with human immunodeficiency virus acquired immunodeficiency syndrome" OR "people living with hiv" OR "HIV Infections"[mesh] OR "HIV infection" OR "HIV infections" OR PLWHIV OR "HIV positive" OR "HIV-positive" OR "HIV+" OR "HIV infected" OR "HIV-infected" OR "HIV seropositivity"[mesh] OR "HIV seropositivity") AND OR Cote d'Ivoire[mesh] OR Ethiopia[mesh] OR Kenya[mesh] OR Malawi[mesh] OR South Africa[mesh] OR Uganda[mesh] OR Zambia[mesh] OR Ivory Coast[tiab] OR Cote d'Ivoire[tiab] OR Ethiopia[tiab] OR Kenya[tiab] OR Malawi[tiab] OR South Africa[tiab] OR Uganda[tiab] OR Zambia[tiab] OR "sub Saharan Africa" OR "sub-saharan Africa" OR Africa[tiab] OR Africa[mesh] OR "sub saharan Africa" OR "developing country" OR "developing countries" OR Cameroon[tiab] OR Central African Republic[tiab] OR Chad[tiab] OR

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Congo[tiab] OR Equatorial Guinea[tiab] OR Gabon[tiab] OR Democratic Republic of the Congo[tiab] OR Burundi[tiab] OR Djibouti[tiab] OR Ethiopia[tiab] OR Kenya[tiab] OR Rwanda[tiab] OR Somalia[tiab] OR Sudan[tiab] OR Tanzania[tiab] OR Uganda[tiab] OR Angola[tiab] OR Botswana[tiab] OR Lesotho[tiab] OR Malawi[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR South Africa[tiab] OR Swaziland[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Benin[tiab] OR Burkina Faso[tiab] OR Cote D'ivoire OR Gambia OR Ghana OR Guinea OR Guinea-Bissau OR Liberia OR Mali OR Mauritania OR Niger[tiab] OR Nigeria[tiab] OR Senegal[tiab] OR Sierra Leone[tiab] OR Togo[tiab] OR Cameroon[mesh] OR Central African Republic[mesh] OR Chad[mesh] OR Congo[mesh] OR Equatorial Guinea[mesh] OR Gabon[mesh] OR Democratic Republic of the Congo[mesh] OR Burundi[mesh] OR Djibouti[mesh] OR Ethiopia[mesh] OR Kenya[mesh] OR Rwanda[mesh] OR Somalia[mesh] OR Sudan[mesh] OR Tanzania[mesh] OR Uganda[mesh] OR Angola[mesh] OR Botswana[mesh] OR Lesotho[mesh] OR Malawi[mesh] OR Mozambique[mesh] OR Namibia[mesh] OR South Africa[mesh] OR Swaziland[mesh] OR Zambia[mesh] OR Zimbabwe[mesh] OR Benin[mesh] OR Burkina Faso[mesh] OR Cote D'ivoire[mesh] OR Gambia[mesh] OR Ghana[mesh] OR Guinea[mesh] OR Guinea-Bissau[mesh] OR Liberia[mesh] OR Mali[mesh] OR Mauritania[mesh] OR Niger[mesh] OR Nigeria[mesh] OR Senegal[mesh] OR Sierra Leone[mesh] OR Togo[mesh] OR "Africa, Central"[Mesh] OR "Africa, Eastern"[Mesh] OR "Africa, Southern"[Mesh] OR "Africa, Western"[Mesh]) AND (Cardiovascular Diseases[mesh] OR Heart Diseases[mesh] OR hypertension[mesh] OR stroke[mesh] OR cardiovascular OR "heart disease" OR hypertension OR "high blood pressure" OR stroke OR "heart attack" OR "Coronary Disease"[Mesh] OR "Cerebrovascular Disorders"[Mesh] OR "Pulmonary Embolism"[Mesh] OR

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"Peripheral Arterial Disease"[Mesh] OR "Peripheral Vascular Diseases"[Mesh] OR "Rheumatic Heart Disease"[Mesh] OR "Venous Thrombosis"[Mesh] OR "coronary disease" OR "pulmonary embolism" OR "cerebrovascular disorder" OR "cerebrovascular disease" OR "peripheral arterial disease" OR "rheumatic heart disease" OR "deep vein thrombosis" OR "ischemic heart disease" OR "heart failure" OR "coronary heart disease" OR "cardiovascular disease" OR "Inflammation"[Mesh] OR inflammation OR inflame* OR "Atherosclerosis"[Mesh] OR atherosclerosis OR "Metabolic Syndrome X"[Mesh] OR "metabolic cardiovascular syndrome" OR "metabolic syndrome" OR "metabolic syndrome X" OR "insulin resistance syndrome X" OR "metabolic X syndrome" OR "cardiovascular biomarker" OR "inflammation biomarker" OR "endothelial function" OR "Interleukin-6"[Mesh] OR IL-6 OR "C-Reactive Protein"[Mesh] OR "C Reactive Protein" OR "C-Reactive Protein" OR "Carotid Intima-Media Thickness"[Mesh] OR "carotid intima media thickness" OR "carotid intima-media thickness" OR "Cholesterol"[Mesh] OR cholesterol OR "Angiography"[Mesh] OR angiography OR "HydroxymethylglutarylCoA Reductase Inhibitors"[Mesh] OR "Hydroxymethylglutaryl CoA Reductase Inhibitors" OR "HMG-CoA Statins" OR "HMG-CoA Reductase Inhibitors" OR "Fluorodeoxyglucose F18"[Mesh] OR 18F-FDG OR "18F FDG" OR "Fluorodeoxyglucose F 18" OR "2-Fluoro-2-deoxy-D-glucose" OR "Venous Thromboembolism"[Mesh] OR "venous thromboembolism" OR cardiometabolic OR "Myocardial Infarction"[Mesh] OR "myocardial infarction" OR "myocardial infarct" OR "Vascular Diseases"[Mesh] OR "vascular disease" OR "Coronary Artery Disease"[Mesh] OR "coronary artery disease" OR "Myocarditis"[Mesh] OR myocarditis OR "Cardiomyopathies"[Mesh] OR cardiomyopathy OR cardiomyopathies OR "cardiac disease" OR "cardiac arrhythmias" OR "Arrhythmias, Cardiac"[Mesh] OR arrhythmia* OR "myocardial

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disease" OR "myocardial diseases" OR myocardiopathy OR myocardiopathies OR carditis OR "dyslipidemias"[MeSH] OR dyslipidemia OR hyperlipidemia OR hypercholesterolemia OR hypertriglyceridemia OR triglyceride OR triglycerides OR HDL OR LDL OR VLDL OR "Lipoproteins, HDL"[Mesh] OR "Lipoproteins, LDL"[Mesh] OR "Lipoproteins, VLDL"[Mesh] OR hyperlipoproteinemia OR lipoprotein(a) OR hyperlipidaemia OR hypercholesterolaemia OR hypertriglyceridaemia OR "Blood Pressure"[MeSH] OR "blood pressure" OR "systolic blood pressure" OR "diastolic blood pressure" OR SBP[tiab] OR DBP[tiab])

Study Selection

Two independent review authors will screen the potentially relevant titles and abstracts according to the pre specified eligibility criteria. Selected articles from the respective databases will be transferred to EndNote 20 (https://endnote.com/). These articles will then be transferred to Rayyan software (https://www.rayyan.ai/) for screening of titles and abstracts. Full text articles will be retrieved after screening the titles and abstracts. Any disagreements arising from record screening will be resolved through discussion.

Data Extraction

A pre-designed data extraction form will be used by two review authors who will independently extract data on prevalence or incidence of any NCD (cardiovascular disease, cancer, mental illness (specifically , depression), chronic respiratory disease and diabetes) among PLHIV, prevalence or incidence of NCD risk factors (hypertension, hyperlipidaemia, dyslipidaemia, physical inactivity, obesity, smoking and pollution) in PLHIV, study design, sample size, participants' age, recruitment methods, study country and date of study publication. The same review authors will independently assess each included study for risk of bias with respect to sequence generation, incomplete outcome data, selective reporting, and other potential sources of bias.

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Discrepancies will be resolved by discussion or by consulting with a third review author. Data will be transferred to Stata version 16 (Stata IC/16.0, StataCorp LLC).

The strength of the body of evidence will be assessed using the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) framework.

Missing Data

Study authors will be contacted concerning missing data on either outcomes or risk of bias.

Data Synthesis

Meta-analysis will be performed using the **"metan"** and **"metaprop"** commands in Stata version 16. A random-effects meta-analysis will be used to estimate pooled prevalence or incidence of the 5 major NCDs among PLHIV. The pooled prevalence or incidence of hypertension, hyperlipidaemia, obesity, overweight and waist circumference in addition to the prevalence of modifiable risk factors (tobacco use, alcohol intake, diet, and physical activity) will be determined. The main outcome of interest will be the prevalence of NCD and NCD risk factors amongst PLHIV with respective 95% confidence intervals. Forest plots will be constructed to display meta-analysis results. Chi-square test and I^2 statistic will be used to measure statistical heterogeneity between studies. If there is significant heterogeneity, subgroup analysis will be used to investigate potential sources; this may include the type of NCD, country, or gender (Male/Female). Publication bias will be assessed using funnel plots and the Stata **"metabias"** command.

As part of the analysis we will compare the systematic review results with Patel 2018[2] and note any differences in the cumulative prevalence/incidence of NCDs among PLHIV post 2016.

The study is scheduled to start in December 2021 and end by June 2022.

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Patient and public Involvement

No patient involved.

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Author Contribution: MM -Conception of the study, Writing - original draft and editing; AM - Conception and design of the study, Writing- reviewing and editing, Supervision. All authors approved the final version to be submitted.

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Section and topic	Item No	Checklist item	Page Number
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	6
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	17
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Not Applicable
Support:			
Sources	5a	Indicate sources of financial or other support for the review	17
Sponsor	5b	Provide name for the review funder and/or sponsor	17
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	Not Applicable
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	4-7
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7-8
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	9-12
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	12-13

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Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	12
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	12
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	13
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8-9
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	13
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	13
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	13
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	13
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	NA
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	13
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	13

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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